

Algorithm for Curve Sketching

1. Determine any restrictions in the domain. State any horizontal and vertical asymptotes or holes in the graph.
2. Determine the intercepts of the graph
3. Determine the critical numbers of the function (where is $f'(x) = 0$ or undefined)
4. Determine the possible points of inflection (where is $f''(x)=0$ or undefined)
5. Create a sign chart that uses the critical numbers, possible points of inflection, and vertical asymptotes as dividing points.
6. Use the sign chart to find intervals of increase/decrease and the intervals of concavity. Use all critical numbers, possible points of inflection, and vertical asymptotes as dividing points.
7. Identify local extrema and points of inflection
8. Sketch the function

Example 1: Use the algorithm for curve sketching to analyze the key features of each of the following functions and to sketch a graph of them.

a) $g(x) = x^3 + 6x^2 + 9x$

1. No restrictions on the domain; no asymptotes

2. x -intercepts:

$$0 = x^3 + 6x^2 + 9x$$

$$0 = x(x^2 + 6x + 9)$$

$$0 = x(x + 3)^2$$

$$x = 0 \text{ or } x = -3$$

$(0, 0)$ and $(-3, 0)$ are x -intercepts

y -intercept:

$$g(0) = 0$$

$(0, 0)$ is the y -intercept

3. $g'(x) = 3x^2 + 12x + 9$

$$0 = 3(x^2 + 4x + 3)$$

$$0 = 3(x + 3)(x + 1)$$

Critical Numbers: $x = -3$ and $x = -1$

Critical Points: $(-3, 0)$ and $(-1, -4)$

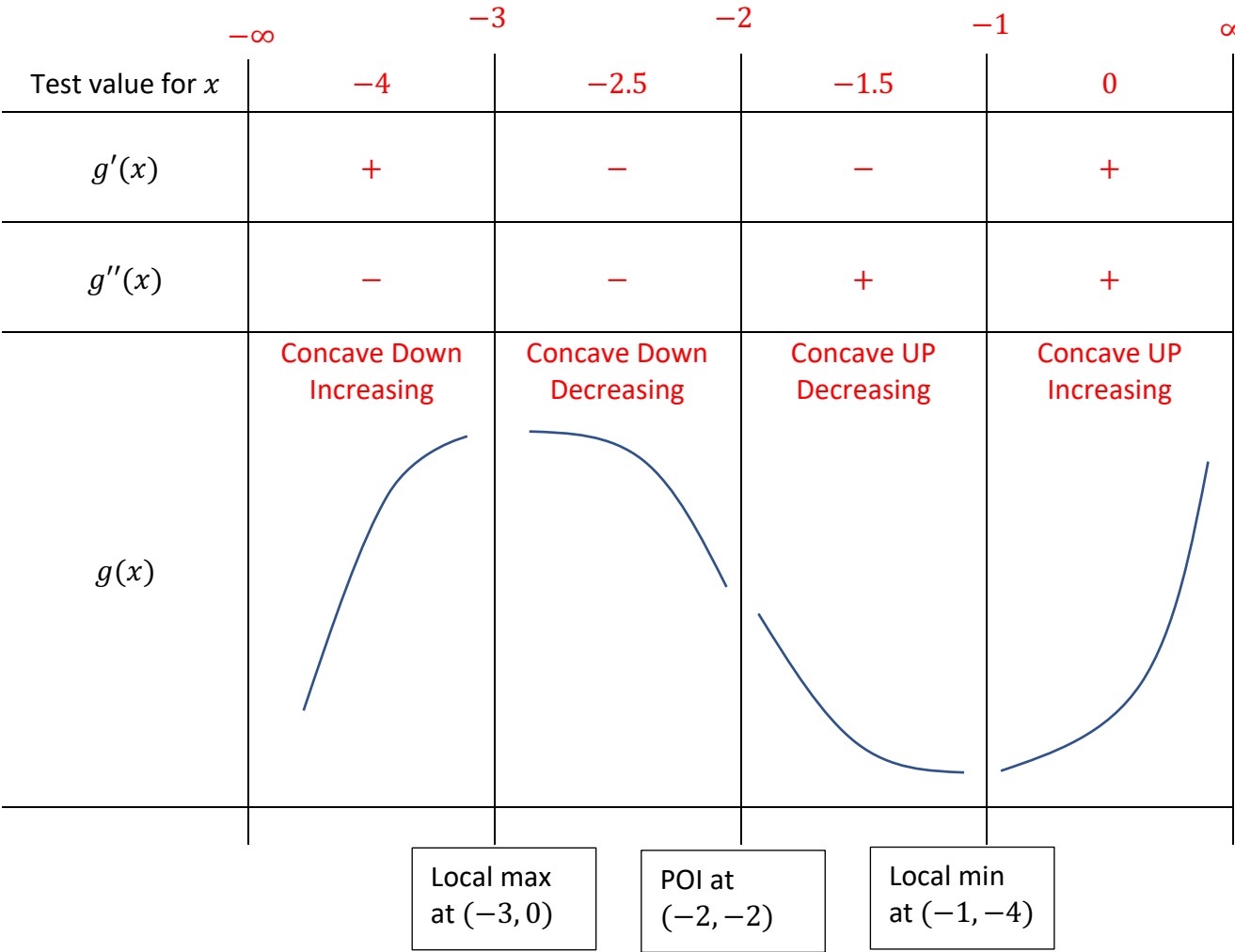
4. $g''(x) = 6x + 12$

$0 = 6(x + 2)$

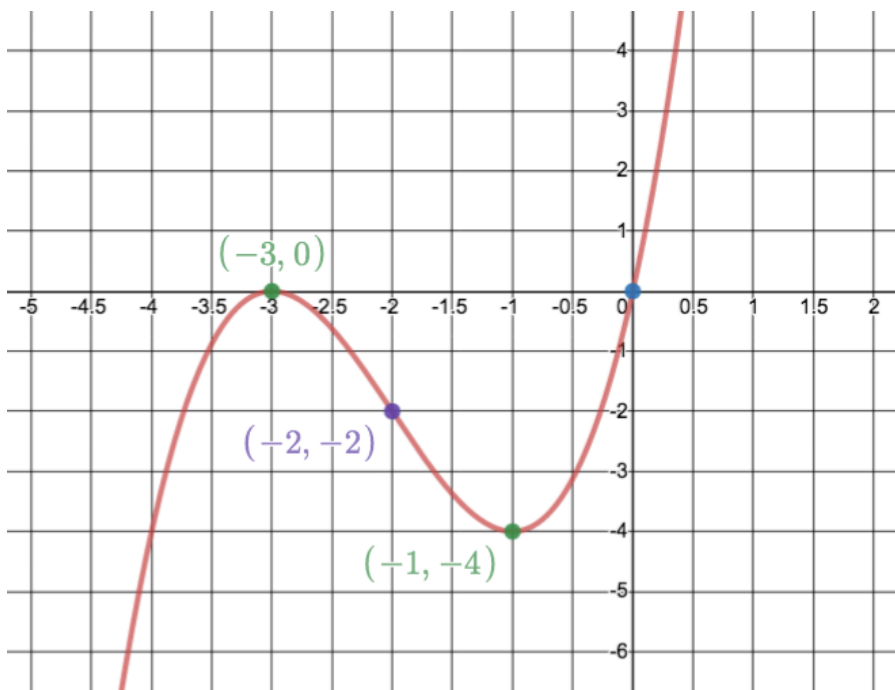
$x = -2$

$(-2, -2)$ is a possible point of inflection

5/6/7.



8.



$$\text{b) } f(x) = \frac{1}{(x+1)(x-4)} = \frac{1}{x^2-3x-4}$$

1. VA: $x = -1$ and $x = 4$ (from restrictions on denominator)

HA: $y = 0$ (degree in denominator is higher than degree in numerator)

2. x-int: NONE

$$\text{y-int: } f(0) = \frac{1}{(0+1)(0-4)} = -\frac{1}{4} \quad (0, -0.25)$$

$$3. f'(x) = \frac{0(x+1)(x-4) - (2x-3)(1)}{(x^2-3x-4)^2}$$

$$f'(x) = \frac{-2x+3}{(x^2-3x-4)^2}$$

Check for zeros:

$$0 = -2x + 3$$

$$x = \frac{3}{2} = 1.5$$

Check for restrictions:

$$0 = (x^2 - 3x - 4)^2$$

$$0 = [(x - 4)(x + 1)]^2$$

$$x = 4 \text{ and } x = -1$$

However, these are not in the domain of $f(x)$, therefore are NOT critical numbers

Critical Number: $x = \frac{3}{2} = 1.5$

Critical Point: $f(1.5) = -0.16$ $(1.5, -0.16)$

$$4. f''(x) = \frac{-2(x^2-3x-4)^2 - 2(x^2-3x-4)(2x-3)(-2x+3)}{(x^2-3x-4)^4}$$

$$f''(x) = \frac{(x^2 - 3x - 4)[-2(x^2 - 3x - 4) - 2(2x - 3)(-2x + 3)]}{(x^2 - 3x - 4)^4}$$

$$f''(x) = \frac{-2(x^2 - 3x - 4) - 2(2x - 3)(-2x + 3)}{(x^2 - 3x - 4)^3}$$

$$f''(x) = \frac{-2x^2 + 6x + 8 - 2(-4x^2 + 12x - 9)}{(x^2 - 3x - 4)^3}$$

$$f''(x) = \frac{-2x^2 + 6x + 8 + 8x^2 - 24x + 18}{(x^2 - 3x - 4)^3}$$

$$f''(x) = \frac{6x^2 - 18x + 26}{(x^2 - 3x - 4)^3}$$

Check for zeros:

$$0 = 6x^2 - 18x + 26$$

$$b^2 - 4ac = (-18)^2 - 4(6)(26) = -300$$

$$b^2 - 4ac < 0 \text{ therefore no real solutions}$$

$f(x)$ has not points of inflection

Check for restrictions:

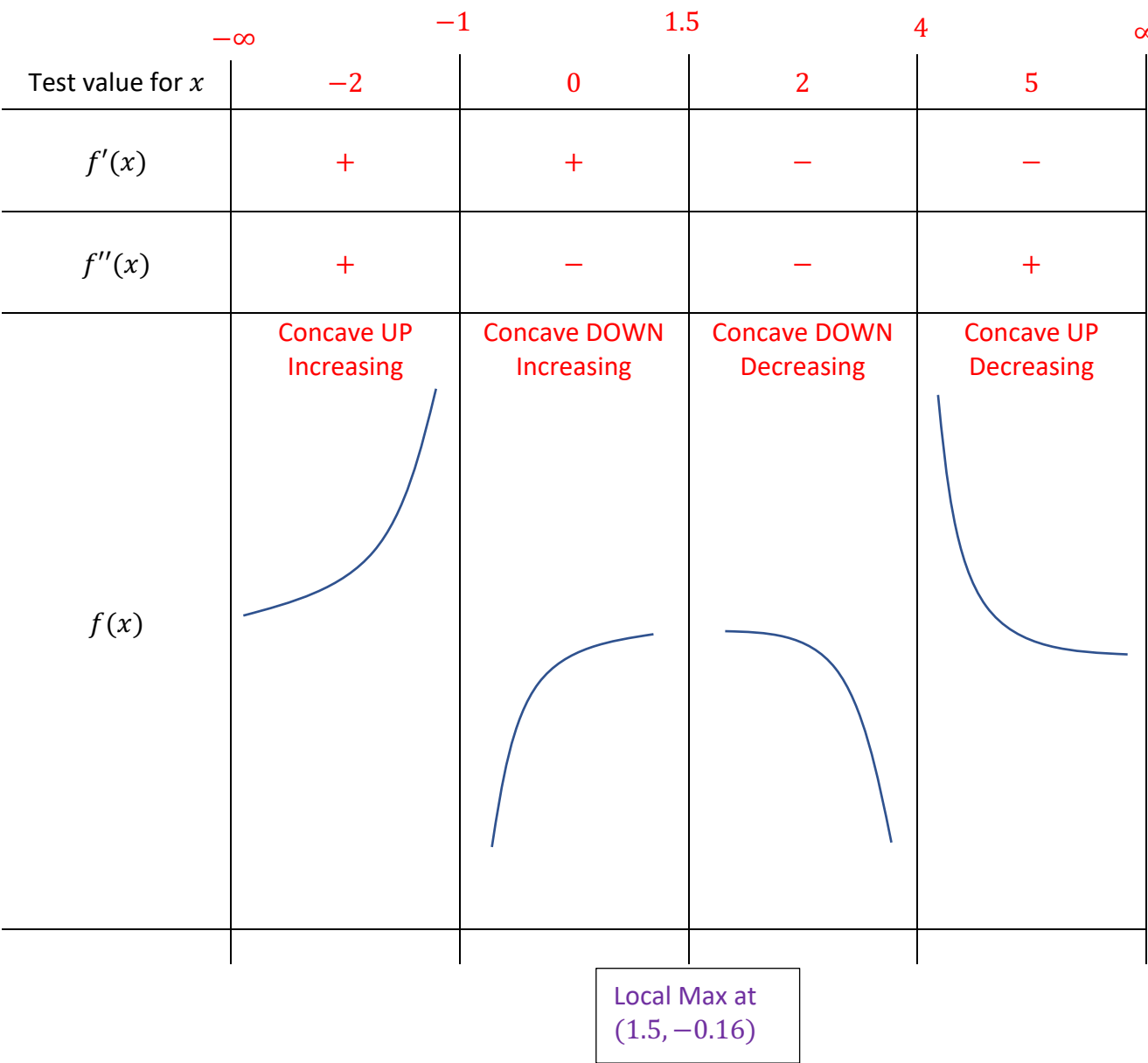
$$0 = (x^2 - 3x - 4)^3$$

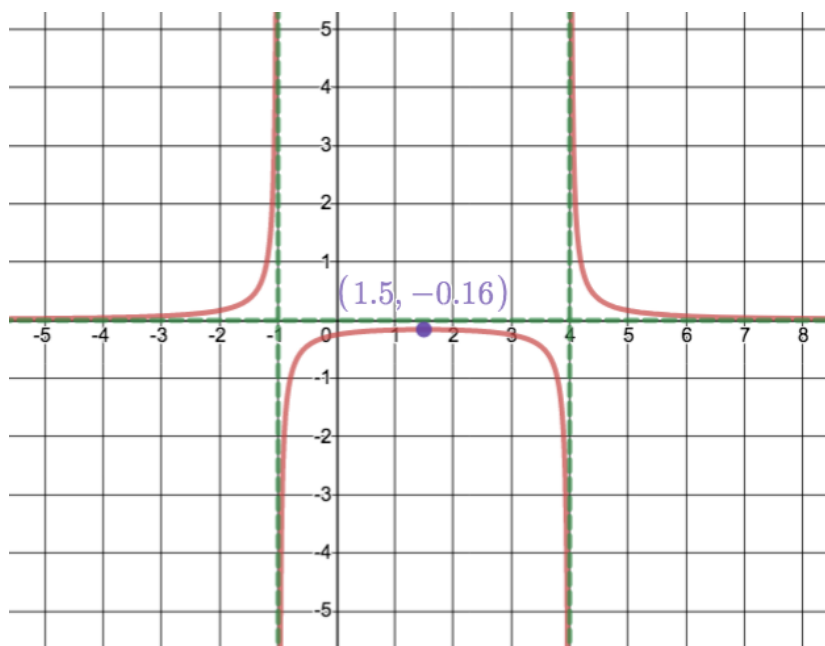
$$0 = [(x - 4)(x + 1)]^3$$

$$x = 4 \text{ and } x = -1$$

Possible changes in concavity at $x = -1$ and $x = 4$ but $f(x)$ is not defined at these values either, therefore they can NOT be considered points of inflection.

5/6/7.





c) $h(x) = x^4 - 5x^3 + x^2 + 21x - 18$

1. No restrictions; no asymptotes

2. x-int:

$$0 = x^4 - 5x^3 + x^2 + 21x - 18$$

$$0 = (x - 1)(x^3 - 4x^2 - 3x + 18)$$

$$0 = x^4 - 5x^3 + x^2 + 21x - 18$$

$$0 = (x - 1)(x^3 - 4x^2 - 3x + 18)$$

$$0 = (x - 1)(x + 2)(x^2 - 6x + 9)$$

$$0 = (x - 1)(x + 2)(x - 3)^2$$

$$x = 1 \quad x = -2 \quad x = 3$$

$$(1, 0), (-2, 0) \text{ and } (3, 0)$$

y-int: $h(0) = -18 \quad (0, -18)$

Factor: $x^4 - 5x^3 + x^2 + 21x - 18$

Possible zeros: $\pm 1, 2, 3, 4, 6, 9, 18$

$h(1) = 0$; $x - 1$ is a factor

1		1	-5	1	21	-18
			1	-4	-3	18
		1	-4	-3	18	0

Factor: $x^3 - 4x^2 - 3x + 18$

Possible zeros: $\pm 1, 2, 3, 4, 6, 9, 18$

$h(-2) = 0$; $x + 2$ is a factor

-2		1	-4	-3	18
			-2	12	-18
		1	-6	9	0

$$3. h'(x) = 4x^3 - 15x^2 + 2x + 21$$

$$0 = 4x^3 - 15x^2 + 2x + 21$$

$$0 = (x + 1)(4x^2 - 19x + 21)$$

$$0 = (x + 1)[4x^2 - 7x - 12x + 21]$$

$$0 = (x + 1)[x(4x - 7) - 3(4x - 7)]$$

$$0 = (x + 1)(4x - 7)(x - 3)$$

$$\text{Critical Numbers: } x = -1, \frac{7}{4}, 3$$

$$\text{Critical Points: } (-1, -32), (1.75, 4.39), \text{ and } (3, 0)$$

$$4. h''(x) = 12x^2 - 30x + 2$$

$$0 = 2(6x^2 - 15x + 1)$$

$$0 = 6x^2 - 15x + 1$$

$$x = \frac{15 \pm \sqrt{(-15)^2 - 4(6)(1)}}{2(6)}$$

$$x = \frac{15 \pm \sqrt{201}}{12}$$

$$x \cong 2.43 \quad \text{and} \quad x \cong 0.07$$

$$\text{Possible points of inflection: } (2.43, 2.05) \text{ and } (0.07, -16.56)$$

$$\text{Factor: } 4x^3 - 15x^2 + 2x + 21$$

$$\text{Possible zeros: } \pm 1, \frac{1}{2}, \frac{1}{4}, 3, \frac{3}{2}, \frac{3}{4}, 7, \frac{7}{2}, \frac{7}{4}, 21, \frac{21}{2}, \frac{21}{4}$$

$$h'(-1) = 0; x + 1 \text{ is a factor}$$

-1	4	-15	2	21
		-4	19	-21
	4	-19	21	0

5/6/7.

Test value for x	$-\infty$	-2	-1	0	0.07	1	1.75	2	2.43	2.5	3	4	∞
$h'(x)$		$-$		$+$		$+$		$-$		$-$		$+$	
$h''(x)$		$+$		$+$		$-$		$-$		$+$		$+$	
$h(x)$	Decreasing Concave UP		Increasing Concave UP		Increasing Concave DOWN		Decreasing Concave DOWN		Decreasing Concave UP		Increasing Concave UP		
													
	Local min at $(-1, -32)$		POI at $(0.07, -16.56)$		Local max at $(1.75, 4.39)$		POI at $(2.43, 2.05)$		Local min at $(3, 0)$				

